



Verified Carbon Standard

100 MW SOLAR PV PROJECT BY ERBL, BANGLADESH

Document Prepared by LGAI Technological Center S.A. (Applus+ Certification)



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Report Title	100 MW Solar PV project by ERBL, Bangladesh
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Summary:

Validation purpose: The primary objective of this project is to reduce greenhouse gas (GHG) emissions through the deployment of Solar Photovoltaic Panels by Energon Renewables (BD) Ltd is special purpose vehicle (SPV) formed by Orion Group. The project involves the establishment of a 100 MW_{AC} grid-tied solar power system, incorporating Half-Cell lower operating temperate Solar PV modules, grid-tied inverters, power transformers, power evacuation infrastructure, energy metering points, switch yards, and associated civil constructions. Notably, the project is situated on a greenfield site where no pre-existing power plants were operational. Energon Renewables (BD) Ltd., referred to as the "project proponent," is overseeing the installation of the entire solar power facility, denoted as the "proposed project activity." This initiative is taking place in Moidhara and Bora Durgapur Village sub-districts under Mongla & Rampal Upazila, Bagerhat District, within the Khulna Division of Bangladesh. The project aims to generate carbon credits under the Verified Carbon Standard (VCS) mechanism. Detailed information about the project activity is outlined below.

S.I No.	Location	Installed Capacity (MW _{AC})	Date of Commissioning
1.	100 MW Solar PV project by ERBL, Bangladesh.	100	29-December-2021

During the 7 years of first crediting period i.e., 29-December-2021 to 28-December-2028, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 92,734 tCO₂e per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 649,135 tCO₂e.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project's monitoring plan is assessed against "ACM0002 Grid-connected electricity generation from renewable sources — Version 21.0"/8/.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline and standard version
- CDM Validation and Verification Standard for project activities, version 03.0/6/
- VCS standard, v4.5/8/

- VCS Program Guide, v4.4/8/

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

A risk-based approach has been followed to perform this validation activity. In the course of validation, 06 Corrective Action requests (CARs) and 05 Clarification Requests (CLs), 00 Forward action request (FARs) were raised and successfully closed (for Validation). The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews

and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus + Certification) has been appointed by “Energon Renewables (BD) Ltd.” to perform the validation of the project entitled “100 MW Solar PV project by ERBL, Bangladesh by Energon Renewables (BD) Ltd.”, VCS standard V 4.5 and VCS Program Guide V4.4/8/. The objective of this validation activity is to have an independent third party for the assessment of the project design, ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project’s baseline is assessed against “ACM0002 Grid-connected electricity generation from renewable sources — Version 21.0”/09/
- The project’s monitoring plan is assessed against “ACM0002 Grid-connected electricity generation from renewable sources — Version 21.0”.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol/06/, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline V4.4 and standard version V4.5/09/.
- CDM Validation and Verification Standard for project activities version 03.0/06/
- VCS standard, v4.5 /8/
- VCS Program Guide, v4.4 /8/

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified certified units (VCUs). This report contains the findings and resolutions from the validation of the project activity.

1.2 Scope and Criteria

The scope of the validation is the independent and objective review of the VCS Project Description. The VCS PD /04/ are reviewed against the relevant criteria (see 1.1) and decisions by the CDM Executive Board and VCS executive board, including the approved baseline and monitoring

methodology. The validation was based on the guidance given in the VCS Program Guide, V4.4 and VCS standard, v4.5.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCU's. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design and monitoring report combined. The only purpose of the validation is its usage during the registration/issuance process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

The validation has been planned and organized to achieve a Reasonable Level of assurance as per the requirement of VCS.

1.4 Summary Description of the Project

The main purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic (SPV) Panels and generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas Projects in Manikganj District of Dhaka region of Bangladesh by Energon Renewables (BD) Ltd. Electricity generated from the project activity is supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity. The project was commissioned on 29-December-2021. The commissioning certificate /01/ and power purchase agreement /09/ is reviewed by the validation team and found to be appropriate. The validation team has validated the equipment 's and monitoring arrangements during the onsite visit and found no deviation. The project activity instance was found to be implemented as it has been defined in the VCS PD. The details of project capacity and location details for all the project instances are as follows: -

S.I. No.	Location	Installed Capacity (MWAC)	Date of Commissioning
1	Nali in Shibaloy, Upazila of Manikganj District, under Dhaka Division of Bangladesh.	100	29-December-2021

During the 7 years of first crediting period i.e., 29-December-2021 to 28-December-2028, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately

92,734 tCO₂e per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 649,135 tonnes of tCO₂e.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation Scope: The scope is defined as an independent and objective review of the project design document. The VCS PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard version 4.5 and guideline version 4.4, including the approved baseline and monitoring methodology AMS-I.B.: Mechanical energy for the user with or without electrical energy – Version 13.0. The validation was based on the requirements in the Validation and Verification Standard for project activities version 03.0, and VCS guideline version 4.4 and standard version 4.5.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project document.

Validation Process: The project assessment is based on the Clean Development Mechanism Validation and Verification Standard for project activities version 03.0 and VCS standard version 4.5 and guideline version 4.4 and is conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed. Once the project is received by the assessment team, the members of the assessment team carried out:

- A desk review of the project design documentation;
- Follow-up interviews with project stakeholders;
- The resolution of outstanding issues and the issuance of the final Validation report and opinion.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI TECHNOLOGICAL CENTER S.A. (APPLUS+ CERTIFICATION) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from validating the identified criteria.

Appointment of the assessment team: -

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The four qualification levels for team members that are assigned by formal appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A) / Auditor in Training (AiT).
- Technical Expert (TE).
- Technical Reviewer (TR).

The sectoral scope / technical area knowledge linked to the applied methodology/ies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Dr. Atul Takarkhede	LA/TE	YES	YES	YES	YES
Dr. N Premjit Singh	TR	YES	YES	YES	NA

2.2 Document Review

The details of the document observed during the validation process are listed below in appendix 1 of this report.

2.3 Interviews

Organization	Name of Persons/Designation	Site	Topics discussed	Team Member
Energion Renewables (BD) Ltd.	Md. Shehridul Hossain (Plant Incharge)	Mongla & Rampal Upazila, Bagerhat District in Khulna Division. State – Bangladesh	Project description, eligibility, baseline, additionality,	Dr. Atul Takarkhede (Team Leader/

	Md. Makig Uddain (Sr. Assistant Engineer)		Project Implementation, O&M activities, LSC etc, VCS PD, and technical specifications	Technical Expert)
	Md. Shamol Ali (Sr. Engineer)			
	Md. Abdul Matin (Assistant Manager)			
Local stakeholders	Jayanta Mistry	Moidhara Bagerhat District in Khulna Division	Local Stakeholder consultation, Project baseline, ongoing communication	
	Dilip Mistry			
	Mrs. Provati Mistry			
	Subroto Das			
	Ram Prasad			
	Mr. Shiv Mama			
Mr. Sandipan Sinha	Project Manager	-	PD preparation calculations, IRR calculations etc.	

2.4 Site Visits

The project involves installation of solar project the host country i.e., Bangladesh, where it will displace electricity from an electricity distribution system that is or would have been supplied by at least one fossil fuel fired generating unit i.e., in the absence of the project activity, of solar project is located at Moidhara and Bora Durgapur Village sub-district under Mongla & Rampal Upazila, Bagerhat District in Khulna Division, country Bangladesh.

Below are the details of Site visits along with date: -

Date of site visit: 21-November-2023

Activity performed on-site	Site location	Date	Team member
• An assessment of the project design and technical specification, project location, implementation status and operation of the project activity as per the VCS PD. • A review of information flows for generating, aggregating and reporting the monitoring parameters. • Interviews with relevant personnel to determine the operational and data collection procedures as per the VCS PD • A review of calculations and assumptions made in determining the GHG data and emission reductions • An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.	Bagerhat District in Khulna Division, Bangladesh	21-November-2023	Dr. Atul Takarkhede (LA/TE)

Validation team confirmed that project is implemented as per VCS PD is followed as per provision of VCS PD and there is no change in technical specifications of the plant since installation.

2.5 Resolution of Findings

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the Appendix 2.

The final VCS PD submitted by project owners serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country:

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project design document	00	02	00
Description of project activity	01	00	00
Start date	00	00	00

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Application of selected baseline and monitoring methodology and selected			
- Applicability of methodology and standardized baseline	01	01	00
- Deviation from methodology	01	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	00	00	00
- Demonstration of additionality	00	02	00
- Emission reductions	00	01	00
- Monitoring plan	01	00	00
- Stakeholder's consultation process	00	00	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting, SDGs- for validation	00	00	00
Others (please specify)-GSC comment	01	00	00
Total	05	06	00

The list of findings and their resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

No FAR raised during the validation process. Please refer Appendix 2 for details.

3 VALIDATION FINDINGS

3.1 Project Details

The main purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic (SPV) Panels and generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas Projects in Manikganj District of Dhaka region of Bangladesh by Energon Renewables (BD) Ltd. Electricity generated from the project activity is supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity.

Audit Type	Period	Program	VVB Name	Number of years
Validation	29-December-2021 to 28-December-2028	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	7 Years (1 st crediting period - Twice renewal)
Total	-	-	-	-

Baseline Scenario: -

The project intends to use the electricity generated from the solar power activity supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity. The 100 MW solar plant unit is located at the Moidhara village of Bagerhat district in the Khulna division of Bangladesh. The project activity displacing electricity from Bangladesh Grid, which is supplied by predominantly fossil fuel plants. The project activity instances would replace equivalent quantum of fossil fuel dominated grid power that's being used prior to the project activity instances. Hence, pre-project scenario and baseline scenario are the same.

The operational life of the installed solar module is 25 years based upon the standard operational and maintenance practices followed at site by PP itself. Same is also verified from the Power Purchase Agreement /09/, EPC (Engineering, procurement, and construction) Agreement /09/ and technical details of the project activity.

The estimated annual average net electricity supplied to the grid is 170,624 MWh and corresponding estimated average emission reduction is 92,734 tCO₂e annually.

The technical specifications/03/ of the equipment's installed for the project activity are as follows:

-

Equipment	Parameter	Unit	Technology Standards		
Solar PV module	Make		V1-1.34 E-W Longi_535/540/545		
	Peak Power	Wp	535	540	545
	Mounting Type		Fixed		
	No. of PV modules		248528		
Inverter	Make & Model		Sungrow SG3125HV-30		
	Rated capacity	kW	3125		
	Power factor		0.980		

During site visit assessment team checked the details of the project proponent are as below:

Organization name	Energion Renewables (BD) Ltd.
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Contact person	Mehedi Islam Aneek
Title	Assistant Vice President
Address	153-154 Tejgaon Industrial Area, Dhaka-1208, Bangladesh
Telephone	+8801714082308, +8801552402427
Email	mehedi.opesl@orion-group.net

assessment team checked the details of the other entity as below:

Organization name	First Climate (India) Pvt. Ltd.
Role in the project	Project Representative
Contact person	Partha Pratim Chaudhuri
Title	Managing Director
Address	Unit 903, ERGO Tower, Plot No. A1/4, Block EP & GP, Sector V, Salt Lake, Kolkata- 700 091
Telephone	+91-98310 12824
Email	partha.chaudhuri@firstclimate.co.in

Project start Date: - Project Start Date: 29-December-2021 which is the date of commissioning of the project activity. Assessment team checked the commissioning details from the commissioning certificates /01/ provided by the project proponent and start date found correct.

Project crediting period: - Assessment team confirms that the crediting period dates for the project is as below: -

Project crediting period: Renewal crediting period

Start date of Crediting period: 29-December-2021

End date of crediting period: 28-December-2021

Total number of years: 07

The project activity adopts renewable crediting period of 07 years which can be renewed for maximum 2 times.

Project scale & estimated GHG Emissions Reductions or Removals: -

Since the installed capacity of the project activity is 100 MW_{AC}. Assessment team confirms that the scale of the project activity as per VCS standard version 4.5, /09/ the projects are classified as follows:

- 1) Projects: Less than or equal to 300,000 tonnes of CO₂e per year
- 2) Large Projects: Greater than 300,000 tonnes of CO₂e per year

The project activity instances being included currently have less than 300,000 tCO₂e Emission reductions, hence these project activity instances are classified as “Projects”

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
<u>29-December-2021 – 31-December-2021 (3 days)</u>	784
<u>01-January-2022 – 31-December-2022</u>	95,449
<u>01-January-2023 – 31-December-2023</u>	93,551
<u>01-January-2024 – 31-December-2024</u>	93,037
<u>01-January-2025 – 31-December-2025</u>	92,525
<u>01-January-2026 – 31-December-2026</u>	92,016
<u>01-January-2027 – 31-December-2027</u>	91,510
<u>01-January-2028 – 28-December-2028 (362 days)</u>	90,263
Total estimated ERs	649,135
Total number of crediting years	07

Average annual ERs

92,734

Condition prior to project initiation: -

Assessment team during the desk review and onsite assessment confirms that the project is an installation of solar photovoltaic by Energon Renewables (BD) Ltd. and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. The baseline as described in section 1.13 of this report will continue to be the baseline in the absence of project activity.

Project compliance with applicable laws, statutes, and another regulatory framework: -

Assessment team confirms that the Project has received necessary approvals for development and commissioning for the proposed Solar PV project from the government of Bangladesh and is in compliance to the local laws and regulations. Assessment team checked the Commissioning certificates, power purchase agreements, Installation report for Solar Plants to confirm the project capacity and its relevant statutory requirements as per the host country regulations.

The project is a voluntary initiative by the project proponent and has not been implemented to meet any local/national laws or regulatory compliances /16/. The project has also received Environmental and Social Impact Assessment (ESIA) by Bangladesh Centre for Advanced Studies. The project activity is in compliance with current laws and regulations and there are no legal and / or regulatory requirements that prevent the project implementation Also, the assessment team has confirmed that there is no such compliance requirement with an emission trading program or any binding limits on GHG emissions for the project activity in Bangladesh (host country) as it is a Least Developed Country (LDC). The project has obtained valid consents, installation and operation from the state nodal agencies and is in compliance with local laws and regulations.

Project ownership: -

Energon Renewables (BD) Ltd. is project proponent (PP) of project activity and they have the legal right to control and operate the project activities.

The project ownership has been checked by the Assessment Team and demonstrated through below supporting documents:

- Power Purchase Agreement (PPA)/ 09/

- Commissioning certificate/01/
- EPC Agreement /09/

Further, individual compliance aspects will be provided for each of the project activity instance to be included in the project activity based on the clearances /09/certificate received for implementation of project instance.

Emission Trading programs & other binding limits: -

Assessment team confirms that the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. Further, Declaration in effect of the same has been submitted by project proponent to audit team and found to be correct. Thus, it is concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Leakage Management for AFOLU project: -

Not applicable to the project activity.

Commercially Sensitive Information: -

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.

Sustainable Development: -

Sustainability is defined by three different parameters: - environmental sustainability, social sustainability, and economic sustainability.

- The project activity utilizes renewable energy source for electricity generation instead of fossil fuel-based electricity generation which would have gaseous, liquid and/or solid effluents /wastes. Thus, the project causes no negative impact on the surrounding environment and contributes to environmental well-being.
- The project activity supporting to alleviate poverty by means of direct and indirect employment generation and improves activities by strengthening of local grid of the state electricity utility. Thus, contributes livelihood to the society.

- The project generates electricity utilizing renewable source i.e. solar. Thus, it will increase the contribution of renewable based power generation in the region and will also help in reducing the demand – supply gap of the respective grid.
- The project activity involves substantial amount of investment, manpower, operational amenities etc. Thus, it contributes to direct and indirect employment generation as per the requirement of skilled and semi-skilled manpower, both during construction and operational phase.

Further, in line with VCS Standard Version 4.5/08/, “The project proponent must demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period, and in each subsequent monitoring period.” As per the provision of implementation deadline, “Projects registered before 20 January 2023 shall demonstrate contributions to at least three SDGs by 20 January 2025”. However, PP have provided the information on the SDG contributions by the project activity for SDGs:

1. SDG 7: Ensure access to affordable, reliable, sustainable, and modern energy for all.
2. SDG 8: Promote sustained, inclusive, and sustainable economic growth, full and productive employment and decent work for all
3. SDG 13.0: Take urgent action to combat climate change and its impacts.

Hence, in view of the assessment of VCS PD /04/ and supporting documents are listed in Appendix 1 of this report. The validation team is able to confirm that the description contained in the VCS PD of the project activity and the technical aspects of its implementation. Consequently, assessment team confirms that the project description of the project contained in the joint VCS PD /04/ to be complete and accurate. The VCS PD complied with relevant forms and guidance for completing the VCS PD.

3.2 Safeguards

3.2.1 No Net Harm

The project activity promotes environmental and socio-economic well-being as it results in zero GHG emissions due to installation and operation of clean, renewable energy technology for electricity generation.

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. Moreover, the stakeholder meeting took place prior to the start date of the project activity which fulfil the requirement VCS Standard 4.5. The VVB checked the relevance of

the dates during the validation Onsite visit. The project owner has organized the stakeholder consultation meeting on 10-August-2018 for all the local people who have direct or indirect connection with this project activity. PP informed nearby localities through notice written in local language and direct invitation as well. The detail of the invitation date and stakeholder meeting date is as below:

Sr. No.	Groups	Local Stakeholder Meeting Date	Venue
1.	Business Group	13-April -2022	House Yard of Sabuj Shekh, Chakgona, Rajnagar Union, Rampal, Bagerhut
2.	Women Group	14-April -2022	House Premises of Renuka Das, Moidhara, Rajnagar Union, Rampal, Bagerhut
3.	Civil Society	15-April -2022	House of AbuBakar, Kaliker Ber, Rajnagar Union, Rampal, Bagerhut
4.	Fishermen Group	16-April -2022	Project Gate, Rajnagar Union, Rampal, Bagerhut

The local stakeholders' consultation meeting was attended by local persons including local villagers, local vendors, and technology suppliers.

The stakeholders identified by the project participant were local people/ NGOs who are the major population of the area, local communities, technology supplier, project proponent representatives, O&M Team and other people involved in the project and above groups. Validation team verified the list of participants who attended the stakeholder meeting and feedback questionnaire and confirms the stakeholders identified are relevant. The validation team also verified the minutes of meeting to note that no negative comments were received and during interviews with the stakeholders found no grievances.

Assessment team found that, there were some suggestions received from respondents & attendees of the LSC meeting. Same were responded by the PP during LSC meeting only. Project proponent has incorporated the same in minutes of meeting and project proponent has bound to fulfilled the grievances for open feedback. Assessment team checked the PP responses with minutes of meeting and also verified it through interviews with the local people during Validation site visit. Assessment team noted that all the concerns raised during LSC meeting were satisfactorily responded by PP. Also during validation site visit, VVB team also visited Moydara village near to power plant and during interaction with stakeholders noted that locals do not have any grievance towards project. Rather locals were happy with the project as PP has carried out some social activities. Also it was noted that Locals and company has regular interactions with each other formally through periodical meetings and informal through local employees & as well as direct interactions.

Moreover, the grievance register is also put onsite for the stakeholder to comment on any grievances during the operation lifetime of the project activity, the grievances from the stakeholder

if found suitable will be addressed immediately by the top management and thus the approach is found appropriate for the project activity.

Assessment team conclude that the stakeholders can anytime lodge their grievances if any in the register over the operational life time of the project and no negative opinion observed during the interview with local people.

3.2.3 Environmental Impact

The ESIA assessment for the project has been prepared by the Bangladesh centre for advanced studies (BCAS). The report was approved by the Ministry of Environment, Government of Bangladesh/14/. The ESIA report for the project covers all aspects of the project including capacity, interaction with other plants in the vicinity, natural resources of the project including management, social and economic impacts, technology and materials used, current land use in the region, any historical or protected site within the project boundaries, geological assessment of the project site and any communities affected by the project.

Thus, the adequacy of the analysis carried out by the project participant for the impact of the implementation of the project activity on the environment has been validated as per the VVS standard for project activity version 03.0/06/.

3.2.4 Public Comments

Summarize any public comments submitted during the public comment period. Assess whether the project proponent has taken due account of all and any comments, and provide an overall conclusion regarding public comments.

Include the project proponent's response to each comment, describe any resultant changes to the project design and provide an explanation of how the project proponent's responses are appropriate. Assessment team noted that this project was open for public comment from 07-November-2023 to 07-December-2023. A comment related to additionality in section 2.4 of the VCS PD document mentioned was received during the public commenting period. As per the CDM tool for common practice analysis (Tool 24 v3.1), para 13, "*Step 1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity*". The capacity of proposed project activity is 100 MW. So applicable range for the analysis is 50 MW to 150 MW. PP has referred "Bangladesh Power Development Board for list of installed projects in

Bangladesh till end of 2020^{1&2}” for the list of project commissioned with capacity. The reports were published by Bangladesh Power Development Board which is government agency operating under the Ministry of Power, Energy and Mineral Resources, Government of the People's Republic of Bangladesh. Thus, information is acceptable to VVB team. The project proponent has subsequently updated the additionality criteria in accordance with Tool 24 and provided a detailed explanation in section 3.5.5. of this report. The VVB (Validation and Verification Body) team has assessed and resolved the mentioned additionality issues as per the updates made by the project proponent.

3.2.5 AFOLU-Specific Safeguards

The project activity is not AFOLU project; Hence this section is not required.

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below.

Methodology: “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”³

Sectoral scopes: 1

The methodology refers to following CDM tools:

- **Tool 07:** Tool to calculate the emission factor for an electricity system - Version 07.0⁴
- **Tool 21:** Tool 01 - Tool for the demonstration and assessment of additionality

¹

https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-55-f1971a327c2aebfd37f6f9a8e723d1fb.pdf (accessed reconfirmed on last date 19-June-2026)

²

https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-11-29-05-22-0dea17e09d8a84e72a63312df6b5bdc6.pdf (accessed reconfirmed on last date 19-June-2026)

³ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

– Version 07.0.0, EB 70 annex 08⁵

• **Tool 27:** “Investment Analysis” Version 8.0 & version 12.0⁶

• **Tool 24** – Common practice - Version 03.1, EB 84 annex 7⁷

3.3.2 Applicability

The applied baseline methodology is justified as it has been demonstrated that the proposed project activity is:

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
4.	1. This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity involves construction and operation of grid-connected solar power (renewable energy) project at a site where no renewable energy power plant was operated prior to the implementation of the project activity and therefore the project activity a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (3.a). Hence the project activity meets the	The project activity is greenfield project. VVB confirms the same is checked during the onsite visit and Thus, fulfil criteria.

⁵ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-01-v7.0.0.pdf>

⁶ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v12.pdf>

⁷ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-24-v1.pdf>

		applicability condition of the methodology.	
5.	2. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	The project activity includes installation of grid connected solar power project without any Battery Energy Storage System (BESS). The electricity generated from the project activity is directly fed to the electricity grid. Therefore, the applicability condition 5 is not applicable/relevant to the project activity.	The project activity is grid connected solar power plant. Hence, no battery is used same checked during the Onsite visit .
6.	3. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave	The project activity involves construction and operation of greenfield grid-connected solar power project and hence complies to the applicability condition	The current project activity is newly installed solar power not hydro power plant checked through supporting documents and onsite visit .

	power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity. (c) In case of Greenfield project activities applicable under paragraph (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the	(6.a) of the methodology. Since the project activity does not include capacity additions, retrofits, rehabilitations or replacements of existing plant/unit, the applicability condition 6.bis not applicable/relevant for the project activity. Although the project is a green field project activity but does not include Battery Energy Storage System (BESS). And therefore, the applicability condition 6.c is not applicable/relevant for the project activity. The project activity does not include Battery Energy Storage System (BESS). And therefore, the applicability condition 6.d is not applicable/relevant for the project activity.	
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		renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the		
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	concerned periods of the monitoring period.		
7.	4. In case of hydro power plants, one of the following conditions shall apply: (a)The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b)The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c)The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d)The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply:	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition “7” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.	Solar power project hence this condition is not applicable same is verified with commissioning certificate.

	(i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: (a) Lower than or equal to 15 MW; and (b) Less than 10 per cent of the total installed capacity of integrated hydro power project.		
8.	5. "In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units,	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity, hence the applicability condition "8" is not applicable/relevant to the project activity as the applicability conditions is related	The project activity is a new grid connected Wind power plant/ unit and not Hydro power plant; therefore, these criteria is not applicable for this project activity. This was cross checked by the VVB from the submitted documents such as commissioning

	with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.”	to hydro power projects.	certificate of the project activity.
9.	6. The methodology is not applicable to: (a)Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b)Biomass fired power plants/units.	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 9” is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or	criterion described in point (a) is not relevant to the project activity. This is a Wind power plant/ unit and not a biomass fired plant, therefore criterion described in point (b) is not applicable to the project activity. This was cross checked by the VVB from the submitted documents such as

		biomass fired power plants/units.	commissioning certificate of the project activity.
10.	7. "In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 10 is not relevant as the same pertains to retrofits, rehabilitation s, replacements, or capacity additions.	The project activity is a new grid connected Wind power plant/ unit and not a retrofits, replacement or capacity additions and therefore this criterion is not applicable to the project activity. This was cross checked by the VVB from the submitted documents such as commissioning certificate of the project activity

For each of the applied methodology's applicability conditions, describe the steps taken to assess conformance of the project with the applicability condition. Provide a conclusion with respect to each applicability condition.

Similarly, where the applied methodology provides the project with a number of tools or modules to choose from, describe the steps taken to assess that the appropriate tool or module has been selected. Provide a conclusion with respect to each selected tool or module.

Provide an overall conclusion regarding the applicability of the methodology, and any tools or modules selected by the project proponent.

3.3.3 Project Boundary

As per ACM0002, version 21 "The spatial extent of the project extends to the project power plant and all power plant connected physically to the electricity system that the project power plant that connected to through the national grid and to be confined to physical, geographical site of renewable generating units. The boundary also extends to the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to." Therefore, the entire Bangladesh grid and all connected power plants have been considered in the project boundary for the proposed VCS project activity. The grid is also part of the project boundary as the solar rooftop units are connected to the grid through power evacuation at common point.

The sources and GHG gases involved for proposed Project activity are as below:

Source		Gas	Included?	Justification/Explanation
Baseline	Grid	CO2	Yes	Main emission source
	connected	CH4	NO	Minor emission source
	electricity generation	N2O	NO	Minor emission source
Project	Greenfield	CO2	NO	No CO2 emissions are emitted from the project
	Solar Power	CH4	NO	Project activity does not emit CH4
	Project activity	N2O	NO	Project activity does not emit N2O

3.3.4 Baseline Scenario

As per ACM0002, version 21, “For project activities that displace grid electricity and fossil fuel fired on- site captive electricity, the baseline emission factor should reflect the emissions intensity of the grid and the captive power plant in the baseline scenario i.e. the weighted average emission factor for the displaced electricity is calculated using values based on the historical, prior three-year ratios of electricity from captive plants and the grid. For new facilities, the most conservative (lowest) of the emission factor for the two power sources should be used.

Hence, the baseline for the project activity is the equivalent amount of power from the Bangladesh grid. The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) (having weightage 75%) and build margin (BM) (having weightage 25%). Further the values for the operating margin (OM) and build margin and combined margin calculations described in the “Tool to calculate the emission factor for an Electricity System”

The combined margin of the BANGLADESH National Grid used for the project activity is as follows

Parameter	Value	Source
$EF_{grid, CM,y}$	0.5435 tCO ₂ e/MWh	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values.

EF _{grid, OM,y}	0.5613 tCO ₂ e/MWh	Calculated as the last 3-year weighted average generation and GHG emission for the year 2019-20, 2020-21 and 2021-22 in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)". The annual value of plant wise generation and operating efficiency is obtained based on Data available from Bangladesh Power Development Board (Annual Report of 2019-2020 ,2020-21, 2021-2022. Data related to emission factor for each fossil fuel type is obtained from "CDM Baseline of Bangladesh" Department of Environment (DoE)
EF _{grid, BM,y}	0.4898 tCO ₂ e/MWh	Calculated in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)" based on Data available from Bangladesh Power Development Board (Annual Report 2021- 22 and Data related to emission factor for each fossil fuel type is obtained from CDM Baseline of Bangladesh, Department of Environment (DoE)

In the absence of the project activity, equivalent amount of electricity would have been drawn from Bangladesh grid. Hence, the baseline of the project activity is the equivalent amount of power from the national grid.

Assessment team conclude that the above-mentioned baseline scenario is in line with the applied methodology "ACM0002 Grid-connected electricity generation from renewable sources --Version 21.0". Same is also reflected in final VCS PD version 02. Thus accepted.

3.3.5 Additionality

For demonstrating additionality under VCS, the project activity is required to undergo the following tests

a) Legal Requirement Test: - based on the available literature on Electricity Market Law in Bangladesh it was confirmed that there are no enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions or other legally binding mandates requiring its implementation, or requiring the implementation of a similar technology/measure that would achieve equivalent levels of GHG emission reductions.

b) Additionality Tests:

As per the applied methodology ACM0002, Version 21.0, additionality of the following project activity is demonstrated and assessed by the latest version of Tool 27, “Investment Analysis” Version 12. The PO has adopted the stepwise approach for demonstrating and assessing the additionality of the project activity as follows:

Project was envisaged for capacity of 100 MWAC in Moidhara village of Bagerhat district, Bangladesh. Currently, Whole project activity is fully commissioned and continuously contributing towards emission reduction. VCS PD for this project activity was open for public comments from 7/11/2023 to 07/12/2023. Start date of the Project is 29-December-2021, which is the commissioning date of whole plant covered under project activity.

The demonstration of additionality for the proposed Project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e., Tool 27, “Investment Analysis” version 12.0, the project proponent shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- Investment barrier
- Technological barrier
- Barrier due to prevailing practice
- Other barriers

The project proponent has identified “investment barrier” as the most relevant barrier faced by the project activity. The investment barrier faced by the project activity consists of barrier due to high capital cost and consequent impact on return.

Investment analysis

Determine whether the proposed project activity is economically or financially less attractive than at least one other alternative. To conduct the investment analysis, use the following sub- steps:

Determine appropriate analysis method and Sub-step 2b (Option III): Apply benchmark analysis

a) Suitability of investment analysis, financial indicator and benchmark:

Project proponent had demonstrated that the financial returns of the proposed VCS project activity would be insufficient to justify the required capital investment as per VCS Standard version 4.5. In the VCS PD, project proponent has adopted a conservative approach to identify the benchmark for the project activity. The project is generating revenue in terms of power generated from the Solar power plant being used for selling to the grid. Thus, simple cost analysis (Option I) is not appropriate. Hence out of 2 options, investment comparison analysis (Option II) benchmark analysis (Option III), benchmark analysis is used for the project activity as per project type and decision-making context. Therefore, the Expected return on equity is considered appropriate benchmark. Accordingly, the post-tax Equity IRR has been considered as the relevant financial indicator for the project activity which is acceptable to the assessment team. Moreover, the financial indicator selected by the project proponent is correct based on the fact that tool do not restrict the project proponent to either use project IRR or Equity IRR. This is under the prerogative

of the project proponent to select appropriate indicator based on his preferences to know the IRR based on his equity investment or debt investment. The same is thus acceptable to the assessment team. Assessment team however checked the Equity IRR calculation and found that input assumptions used for the calculation of Equity IRR are applicable at the time of investment decision of the project and thus is in accordance with the relevant guideline of the tool.

“In situations where an investment analysis is carried out in nominal terms and the available IRR benchmarks are in real terms, project proponent shall convert the real term values of benchmarks to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used”

The investment analysis has been carried out in Nominal terms. Accordingly, default value has been adjusted by adding suitable forecasted inflation rate taken from IMF, International monetary Fund⁸). Project Participant has calculated Benchmark based on average of Inflation rate Average Consumer Prices published IMF for the next five years after the start of the project i.e., 13/07/2020, which is the date of EPC contract signing date in line with definition of “Start date”⁹ as per CDM Glossary version 11.0¹⁰. The inflation forecast should be for the duration of the crediting period.

Inflation forecast (Average consumer price) as per IMF for 5 years: 7.22%

VVB observed that the decision to invest in project was done in July 2020 and during this period the applicable version of Investment Analysis was version 10. The default value of expected return on equity was 12.60%. However, at time of Validation of PA, the investment analysis tool, version 12 is applicable to project activity, in which default value of return of equity is mentioned as 11.73%. Therefore, 11.73% expected return on equity is considered to be conservative value. Thus, accepted by the assessment team.

As per the Investment analysis tool 27, version 12.0, default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.73%

Therefore, the Required return on equity (benchmark) was computed in the following manner:
Nominal Benchmark = $\{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$

Where:

Default value for Real Benchmark = 11.73% (as per Appendix of CDM TOOL 27) Inflation Rate forecast for Bangladesh.

⁸ [World Economic Outlook \(October 2023\) - Inflation rate, average consumer prices \(imf.org\)](https://www.imf.org/en/Publications/WEO/Issues/2023/01/18/world-economic-outlook-october-2023)

⁹ *The date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility.*

¹⁰ https://cdm.unfccc.int/Reference/Guidclarif/glos_CDM.pdf

Benchmark estimation:

Benchmark

Description	Value	Sources Link
Default Value for Bangladesh as per UNFCCC guidelines	11.73%	https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v12.pdf
Inflation forecast	7.22%	https://www.imf.org/external/datamapper/PCPIPCH@WEO/BGD?year=2023 Calculated as average of 05 years for the period 2020 to 2024
Benchmark	19.80%	Calculated

b) Parameters and assumptions used:

The project activity is a renewable source of electricity generation and supplies the electricity to the Bangladesh Electricity grid. The key parameters which determine the Equity IRR of the project activity are project cost, PLF and profitability estimates.

In the revised VCS PD, the project cost is based on the Information Memorandum 2020 for financial Part and Feasibility Report for Technical Part of the project activity. The details of the DPR are as below:

Assessment of Project cost: -

Site Name	Project Proponent	Project Capacity (MWac)	Project Cost (In USD Million)	Source
100 MW Solar PV project by ERBL	Energion Renewables (BD) Ltd.	100 MWac	200.28	Information Memorandum 2020

Information Memorandum has been submitted to verification team. The Information Memorandum was available during decision making and financial profitability of the project was decided based on this same document. Verification team checked the Information Memorandum of the project activity and found that consideration of the project cost in revised VCS PD is correct and it is in line with Appendix of Methodological tool 27: "Investment Analysis" Version 12.0 as well as in compliance to VCS Standard. Hence, the project cost consideration is justified. Assessment team checked the actual project cost and still the project do not breach the benchmark. The sensitivity analysis below confirms the same. Since the actual cost is considered, there is no way the cost can go up and thus the same is assessed to be correct.

The profitability of the project, which forms the basis for IRR calculation is based on installed capacity, PLF, tariff rate, O&M cost, depreciation, and taxation.

Assessment of Plant Load Factor (PLF):

PP considered the Plant load factor as per calculation in PVSyst Report: -

Site Name	Project Proponent	Project Capacity (MW)	PLF (%) = 3 rd party report	Source
100 MW Solar PV project by ERBL	Energion Renewables (BD) Ltd.	100 MWac	20.05	PV Syst report, 22/11/2019

PP has considered PV Syst report issued by Sgurr Energy India Pvt. Ltd. Dated 22/11/2019, which is 3rd party entity. VVB team found the same acceptable as in line with the EB 48 Annex 11” *Guidelines for the Reporting and Validation of Plant Load factors*” version 1.0.

Assessment of Electricity Tariff:

Site Name	Project Proponent	Project Capacity (MWac)	Tariff Rate (USD /MWh)	Source
100 MW Solar PV project by ERBL	Energion Renewables (BD) Ltd.	100 MW	138	Power Purchase Agreement dated 25/02/2019

Verification team assessed the tariff and found that same value was available during decision making as Power Purchase Agreement was already signed between *Energion Renewables (BD) Ltd.*, and *Bangladesh Power Development Board (BPDB)* and in conformity with guidance Appendix of Methodological tool “Investment Analysis” Version 12.0. Furthermore, assessment team has also checked the actual tariff as mentioned in sample electricity invoices. IRR is still below benchmark with the consideration of power purchase agreement /09/ signed which is valid for total operational lifetime of the project.

Assessment of O&M Cost:

Site Name	Project Proponent	Project Capacity (MWac)	O&M cost (USD Mn)	Source
100 MW Solar PV project by ERBL	Energion Renewables (BD) Ltd.	100 MW	3.20	Information Memorandum 2020

According to Appendix of Methodological tool “Investment Analysis” Version 12, the cost should be based on the input parameters available at the time of decision making and the PP has submitted Information Memorandum 2020 (incorporating cost of initial year) supporting this consideration. Therefore, considering the above assessment, verification team concluded that the O&M cost considered from respective document in the computation of financial indicator is in conformity with guidance Appendix of Methodological tool “Investment Analysis” Version 12.

C) Cross checking parameter

Name of the parameter	VVB assessment
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Project Cost	The details for the proposed project activity are given below:																		
	Project Developer	Project Capacity (MWac)	Project Cost (In USD Million)																
	Energion Renewables (BD) Ltd.	100	200.28 million																
			2.00 million/MW																
The project cost has been considered from Information Memorandum 2020 along with EPC Contract as both documents were available at the time decision made for the project activity. The assessment team has also checked the actual cost of each project site from the EPC contract and found that, the reduction in project cost is within 10% range of sensitivity analysis. Since the comparison is done with actual project cost and hence, increase of the same in future is not possible. Thus, the project activity is additional with actual project cost. The assessment team also checked the state tariff orders and found that project cost considered for the project is found to be appropriate. Based on the title report "Indicative Tariff for Utility-Scale Solar IPP in Bangladesh by Center for Renewable Energy Services Ltd. (CRESL)" the indicative cost is 1.34 million USD/MW(AC) as against the unit project cost of 2.00 million USD/MW(AC) for the current project activity. The cost difference of 50% owes to higher land development cost as the project is developed. Recent PDD registered from Bangladesh has used 1.9 million USD/MW(AC) for estimation of return- https://registry.terra.org/app/projectDetail/VCS/10045 Based on sectoral scope expert and local knowledge, the project cost considered as per information Memorandum and EPC contract for the proposed project activity is found to be appropriate for solar projects. Also, since the actual cost is available to assessment team and IRR is still within benchmark and thus the same is acceptable. The IRR as per the assumption from the DPR is as follows: <table> <tr> <th>Project Capacity (MW)</th><th>Project Cost (In Million)</th><th>IRR</th><th>Benchmark</th></tr> <tr> <td>100MW</td><td>200.28 million</td><td>8.35%</td><td>19.80%</td></tr> </table> The IRR as per the actual project cost is defined as below: <table> <tr> <th>Project Capacity (MW)</th><th>Actual project cost as per the EPC Contract</th><th>IRR</th><th>Benchmark</th></tr> <tr> <td>100 MW</td><td>200.28 million USD</td><td>8.35%</td><td>19.80%</td></tr> </table>				Project Capacity (MW)	Project Cost (In Million)	IRR	Benchmark	100MW	200.28 million	8.35%	19.80%	Project Capacity (MW)	Actual project cost as per the EPC Contract	IRR	Benchmark	100 MW	200.28 million USD	8.35%	19.80%
Project Capacity (MW)	Project Cost (In Million)	IRR	Benchmark																
100MW	200.28 million	8.35%	19.80%																
Project Capacity (MW)	Actual project cost as per the EPC Contract	IRR	Benchmark																
100 MW	200.28 million USD	8.35%	19.80%																

	As described above actual project cost with benchmark, the project is still additional. Since the comparison is done with actual project cost, the increase of the same in future is not possible. Thus, assessment team is of the opinion that project is still additional with the consideration of actual project cost for the project activity.																																
O&M cost and Escalation in the operational expense = 2.19%	The details for the proposed project activity are given below: <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>O&M Cost (In Million/per MW)</th></tr><tr><td>Energon Renewables (BD) Ltd.</td><td>100 MW</td><td>0.2975</td></tr></table> The O&M cost has been considered from DPR and was available at the time decision made for the project activity. The assessment team has also checked the actual O&M contract for the project activity and found the changes in O&M cost is within threshold limit. Thus, the project activity is additional with actual O&M cost. <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>O&M Cost (In Million/per MW) (Without tax) - Actual</th></tr><tr><td>Energon Renewables (BD) Ltd.</td><td>100 MW</td><td>0.3494</td></tr></table> The assessment team also checked the Bangladesh tariff orders and found that O&M cost and its escalation considered for project is found to be appropriate. IRR value as per the assumptions from the DPR is as below: <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>O&M Cost (In USD Million)</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Energon Renewables (BD) Ltd.</td><td>100 MW</td><td>3.20</td><td>8.35 %</td><td>19.80%</td></tr></table> IRR value based on the actual O&M agreements signed is as below: <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>O&M Cost (In Million/per MW)</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Energon Renewables</td><td>100 MW</td><td>3.41</td><td>8.01%</td><td>19.80%</td></tr></table>	Project Developer	Project Capacity (MW)	O&M Cost (In Million/per MW)	Energon Renewables (BD) Ltd.	100 MW	0.2975	Project Developer	Project Capacity (MW)	O&M Cost (In Million/per MW) (Without tax) - Actual	Energon Renewables (BD) Ltd.	100 MW	0.3494	Project Developer	Project Capacity (MW)	O&M Cost (In USD Million)	IRR	Benchmark	Energon Renewables (BD) Ltd.	100 MW	3.20	8.35 %	19.80%	Project Developer	Project Capacity (MW)	O&M Cost (In Million/per MW)	IRR	Benchmark	Energon Renewables	100 MW	3.41	8.01%	19.80%
Project Developer	Project Capacity (MW)	O&M Cost (In Million/per MW)																															
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Energon Renewables	100 MW	3.41	8.01%	19.80%																													

	(BD) Ltd.																		
	Benchmark for the project as described above along with actual O&M value, the project is still additional. Based on sectoral scope expert and local knowledge, the project O&M cost and its escalation considered as per O&M agreement for the proposed project activity is found to be appropriate for solar projects. Also, the O&M cost is available to assessment team and IRR is still within benchmark, thus, the same is acceptable.																		
Tariff	The details for the proposed project activity are given below: - <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>Tariff Rate (as per PPA)</th></tr><tr><td>Energion Renewables (BD) Ltd.</td><td>100 MW</td><td>138 USD/MWh</td></tr></table> The Tariff rate has been considered from PPA and the same was available at the time decision made for the project activity. The assessment team has also checked the actual PPA for each project site and found there are no changes in tariff rate and is within threshold limit. Thus, the project activity is additional with actual Tariff rate. The tariff considered is fixed tariff and hence there is no any escalation. This is found to be appropriate and accepted. IRR value as per the assumptions from the PPA is as below: <table><tr><th>Project Developer</th><th>Tariff Rate (as per PPA)</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Energion Renewables (BD) Ltd.</td><td>0.1380 USD/KWH</td><td>8.01%</td><td>19.80%</td></tr></table> Since the IRR is still below benchmark with the consideration of Actual Power purchase agreement signed which is valid for total operational lifetime of the project, assessment team confirms that the project is still additional with actual Tariff rate. Moreover, in the above tables the sample invoices of utility are checked and it is confirmed that PPA signed with the rate as mentioned in the Tariff order and hence increase of the same is not possible. Based on sectoral scope expert and local knowledge, the project tariff rate considered as per state tariff order for the proposed project activity is found to be appropriate for solar projects.					Project Developer	Project Capacity (MW)	Tariff Rate (as per PPA)	Energion Renewables (BD) Ltd.	100 MW	138 USD/MWh	Project Developer	Tariff Rate (as per PPA)	IRR	Benchmark	Energion Renewables (BD) Ltd.	0.1380 USD/KWH	8.01%	19.80%
Project Developer	Project Capacity (MW)	Tariff Rate (as per PPA)																	
Energion Renewables (BD) Ltd.	100 MW	138 USD/MWh																	
Project Developer	Tariff Rate (as per PPA)	IRR	Benchmark																
Energion Renewables (BD) Ltd.	0.1380 USD/KWH	8.01%	19.80%																
PLF	The details for the proposed project activity are given below: <table><tr><th>Project Developer</th><th>Project Capacity (MW)</th><th>PLF (%)</th></tr><tr><td>Energion Renewables (BD) Ltd.</td><td>100 MW</td><td>20.05%</td></tr></table> Validation team assessed the PVSyst report prepared by Sgurr Energy Pvt Ltd					Project Developer	Project Capacity (MW)	PLF (%)	Energion Renewables (BD) Ltd.	100 MW	20.05%								
Project Developer	Project Capacity (MW)	PLF (%)																	
Energion Renewables (BD) Ltd.	100 MW	20.05%																	

(3rd Party Entity). Same report has been used in the financials and the emission reduction calculation. PLF estimation by 3rd party engineering company is in line with Para 3 (b) Annex 11, EB 48 and acceptable to the assessment team.

The PLF has been taken from the DPR, and the same has been checked and found that PLF considered for the project activity is within the range of sensitivity analysis and found to be appropriate.

Based on sectoral scope expert and local knowledge, the project PLF considered as per DPR for the proposed project activity is found to be appropriate for solar projects.

IRR for PLF value as per the PVSyst Report = 3rd party DPR,

Guidelines for the reporting and validation of Plant load factor (Annex 11 EB 48)

Project Developer	Project Capacity (MW)	PLF (%)	IRR	Benchmark
Energion Renewables (BD) Ltd.	100 MW	20.05%	8.01%	19.80%

IRR as per the PLF value Based on actual generation for period from December 2022 to November 2023 are given below:

Project Developer	Project Capacity (MW)	PLF (%) - Tariff order	IRR	Benchmark
Energion Renewables (BD) Ltd.	100 MW	18%	8.37%	19.80%

Assessment team also confirms that, by considering the PLF value as per actual generation, the IRR is still below the benchmark, hence, the project is additional.

Tax Rates	Income tax rate (%) for first 15 year	0 % https://www.undp.org/sites/g/files/zskgke326/f_iles/migration/bd/2018.11.11-SREPGen_Report-on-indicative-tariff-for-solar-IPPs.docx.pdf
	Income tax rate (%) after 15 years	37.50% https://www.undp.org/sites/g/files/zskgke326/f_iles/migration/bd/2018.11.11-SREPGen_Report-on-indicative-tariff-for-solar-IPPs.docx.pdf
The above table shows the tax rate considered for the project activity and the same is found suitable. The tax rates assumed corresponds to the tax rate prevailing at the time of taking decision. Hence, these assumptions are appropriate during decision making context and assessment team.		

Sensitivity analysis:

The Guidance on Investment analysis requires the robustness of the conclusion arrived at to be proved through a sensitivity analysis by varying the critical assumptions to a reasonable variation. The project developer has identified Plant Load Factor (PLF), Project cost, tariff rate and O&M cost as critical assumptions. These critical parameters constitute more than 20% of either total project costs or total project revenues. The sensitivity analysis reveals that even under more favorable conditions, the IRR would not cross the benchmark return as given in the following table:

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	5.31%	8.35%	11.62%	32.17%
O&M	8.85%	8.35%	7.84%	-214.00%
Project Cost	11.25%	8.35%	6.13%	-28.43%
Tariff Rate	5.31%	8.35%	11.62%	32.17%

The results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost, PLF and Tariff Rate, Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favorable conditions.

Assessment team also confirmed the breaching values for individual parameters (=Individual Project Developers) and thus confirms that the project is still additional

Project Developer	Project Capacity (MW)	PLF in PVSyst Report	Tariff order PLF	Variation in PLF	Breaching Value for PLF
Energion Renewables (BD) Ltd.	100 MW	20.05%	20.37%	+1.60%	32.17%

Project Developer	Project Capacity (MW)	Project Cost	Actual Cost from EPC	Variation in project cost	Breaching value for Project Cost
Energion Renewables (BD) Ltd.	100 MW	200.28	200.28	0%	-28.43%

Project Developer	Project Capacity (MW)	PPA Tariff USD/MWh	Invoices Tariff USD/MWh	Variation in Tariff	Breaching value in Tariff Rate
Energion Renewables (BD) Ltd.	100 MW	138	138	0%	32.17%

Project Developer	Project Capacity (MW)	Estimated O&M cost (USD million)	Actual O&M cost (USD million)	Variation in O&M	Breaching value in O&M
Energion Renewables (BD) Ltd	100 MW	3.20	3.41	+6.56%	-214.00%

The above discussions show that solar power development is not a financially attractive; hence the project activity is additional and the assessment team considers the approach and calculations acceptable as per the requirements in the methodological tool.

Common practice Analysis:

In accordance with paragraph 57 of the “Tool for the demonstration and assessment of additionality” – (Version 07.0.0) the project has been subjected to an analysis of the extent to which the proposed project type (e.g. technology or practice) has already diffused in the relevant sector and region.” The project meets the criteria of tool “Common Practice”, Version 3.1 as follows:

- **Applicable geographical area:** As confirmed from the commissioning certificate, the state of Bangladesh is considered as applicable geographical area.

As per the requirement of paragraph 9 of the tool “Common Practice”, Version 3.1 “the applicable geographical area should be the entire host country. If the project participants opt to limit the applicable geographical area to a specific geographical area (such as province, region, etc.) within the host country, then they shall provide justification on the essential distinction between the identified specific geographical area and rest of the host country.”

- **Measure:** Project has been confirmed to be renewable energy generation activity, which has also been confirmed through the commissioning certificate and power purchase agreement.

- **Technology & Output:** As confirmed from the commissioning certificate, the Project Developer is generating power using solar panels. The power generated by solar panels is fed to the Bangladesh Electricity Grid.

As According to the methodological tool “Common Practice”, Version 3.1 the step wise demonstration of common practice analysis is verified as follows:

Step 1: - Calculate applicable capacity or output range as +/-50% of the design capacity or output of the proposed project activity:

The size of the project activity is 100 MWac, so the applicable output range as defined in the step is +/-50% of the capacity of the proposed project activity. This output range will come in a capacity range of 50 MW to 150 MW

Step 2: - Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is located Bangladesh therefore, projects in the geographical area of Bangladesh have been chosen for analysis.
- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project is solar. Hence, only solar energy projects have been considered for analysis.
- The project produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 50 MW to 150 MW.
- As per the methodological tool the projects that have started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, are to be selected for consideration. The start date mentioned in PD is

13/07/2020 (Date of EPC contract signing) that resembles to start date definition of CDM project activity which is “the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system), for the CDM project activity or CPA. Where a contract is signed for such expenditures.

Findings of analysis of Step 2

No Similar projects identified, which fulfil above-mentioned conditions have been identified from the annual report of Bangladesh Power Development Board¹¹. Hence $N_{solar} = 0$

Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}

No Projects are identified in Step 2 hence $N_{all} = 0$

Step 4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

No Projects are identified in Step 2 hence $N_{diff} = 0$

Step 5: calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

$$F = 1 - 0/0 = 1$$

$$N_{all} - N_{diff} = 0 - 0 = 0$$

Since the value of factor $N_{all} - N_{diff}$ for the proposed project activity is 0 which is less than 3, the project activity is not a “common practice” within sector in the applicable geographical area.

3.3.6 GHG Emission Reductions and Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline Emissions: As per para 47 of ACM0002, version 21.: ACM0002 Grid-connected electricity generation from renewable sources —Version 21.0/09/, the Baseline emissions for other systems are the product of amount electricity displaced with the electricity produced by the renewable generating unit and an emission factor.

$$BE_y = EG_{PJ,y} \times EF_{grid\ CM,y}$$

where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

¹¹https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-11-29-05-22-0dea17e09d8a84e72a63312df6b5bdc6.pdf

$EF_{grid,CM,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using TOOL07 (t CO2/MWh)
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Since the project is a green-field power plant therefore calculation of quantity of net electricity generation ($EG_{PJ,y}$) is carried out in accordance to section 5.5.1.1, paragraph 49 of the approved consolidated Methodology ACM0002, V 21.0, :

$$EG_{PJ,y} = EG_{facility,y} \text{-----Equation (12)}$$

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Project Emissions: -

The project emission calculation as per para 100 of ACM0002 version 21

For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF} + PE_{GP} + PE_{HP,y} + PE_{BESS,y} \text{ Equation (1)}$$

Where:

PE_y Project emissions in year y (t CO2e/yr)

PE_{FF} Project emissions from fossil fuel consumption in year y (t CO2e/yr)

PE_{GP} Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (t CO2e/yr)

$PE_{HP,y}$: Project emissions from water reservoirs of hydro power plants in year y (t CO2e/yr)

$PE_{BESS,y}$: Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO2e/yr)

The project activity is a solar power project without storage system, and in accordance to ACM0002 version 21, and does not include any fossil fuel consumption, therefore in accordance to Para 35, $PE_y = 0$

Leakage Emissions: - As per the methodology ACM0002 v21, para 61, no other leakage emissions are considered. In accordance to the methodology “The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected”.

Therefore, the net emission reduction is calculated as

$$ER_y = BE_y - PE_y$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 1 29-December-2021 – 31-December-2021 (3 days)	784	0	0	784
Year 1 (2022)	95449	0	0	95449
Year 2 (2023)	93551	0	0	93551
Year 3 (2024)	93037	0	0	93037
Year 4 (2025)	92525	0	0	92525
Year 5 (2026)	92016	0	0	92016
Year 6 (2027)	91510	0	0	91510
Year 7 (2027) 01-January-2028 – 28-December-2028	90263			90263
TOTAL	649,135	0	0	649,135

3.3.7 Methodology Deviations

Assessment team confirms that no methodology deviation is applicable for the present project activity.

3.3.8 Monitoring Plan

Assessment team checked the monitoring practice onsite and checked the guideline of respective State electricity regulatory commission. The detail analysis is as below:

Parameters determined ex-ante:

Baseline emission factor of Bangladesh Grid is established ex-ante based on Tool to calculate the grid emission factor, using a combined approach consisting 75 % operating margin and 25 % build margin. The emission coefficient from official data published in Clean Development Mechanism (CDM) Baseline, Power and Energy Sector, published by Department of Environment, Ministry of Environment and Forest, Bangladesh and Annual Report published by “Bangladesh Power Development Board. Database available to the project participant at the time of submission of VCS PD for validation and global stakeholder’s consultation process. The assumption was verified by the validation team and found to be correct. Provide an overall conclusion regarding the adherence of the monitoring plan to the requirements of the applied methodology and any referenced tools.

Data and Parameters	Unit	Ex-ante Determined Value
Operating margin (OM)	tCO2e/MWh	0.5613
Build margin (BM)	tCO2e/MWh	0.4898
Weighting of operating margin emissions factor (WOM)	-	0.75
Weighting of build margin emissions factor (WBM)	-	0.25
Combined emission factor (CM)	tCO2e/MWh	0.5435

The parameters monitored ex-post:

Parameters	Description	Measurement method and QA/QC procedures	Assessment conclusion
EG _{facility,y}	Quantity of net electricity generation supplied by the project to the Grid in year y.	The parameter would be calculated as: $EG_{facility,y} = EG_{export,y} - EG_{import,y}$ Monitored by the bi-directional main meter and backup meter installed at 132 KV line with accuracy class of 0.2s meters. The readings of the electricity meters will be continuously measured and at least monthly recorded. The data will be cross checked with sales receipts.	Consistent with methodology

Monitoring Arrangement:

To ensure the implementation of project monitoring, PP installed two monitoring meters main and check with 0.2s accuracy on the 132kV unit transformer side of the project site. The assessment team validated the energy meter onsite.

Organizational structure and responsibilities:

The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the online visit.

Data Measurement & Data collection and archiving:

As per the VCS PD, the meter reading records of the main meter will be used for emission reduction calculation and the invoices issued by the grid company will be used for cross-checking. generation data routinely monitored and recorded on a daily basis, with the logbook being updated using Distributed Control System (DCS) data. The VVB will use the same source for verification of emission reductions. The calibration of this meters is in authority of Bangladesh grid once in a year. The assessment team therefore is of the opinion that the PP is capable of implementing the monitoring plan in the context of the project activity.

QA/QC Procedures:

Energy meters with an accuracy of 0.2s are installed at the project site. Calibration of all meters is performed annually by an appropriately qualified third party in accordance with industry standards. Export and import power will be measured continuously using the above-mentioned meters installed on the project site. The meter readings will be read by trained staff and confirmed by the project manager and then saved. Invoices from the grid company can be used to cross-check meter readings for the project activity. It is reported that the data will be kept for 2 years following the end of the crediting period.

It is in line with the BERC notification, 2019/09/ (after PPA signing) the meter will be tested for accuracy at the interval of 1 year. In line with the Gazette Notification by Bangladesh Energy Regulatory Commission (BERC) dated 31 December, 2019¹², the calibration of metering system is

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http://berc.portal.gov.bd/sites/default/files/files/berc.portal.gov.bd/page/a250b6fc_8bcf_4c96_bb20_3c3de2_30467a/2020-03-01-17-04-6c286949cb014aac31754ffa3e86bac2.pdf

required to be carried out at intervals after 5 years (Section 13.6.10), along with periodical yearly testing of meter and associated equipment's.

However, with reference to PPA signed/14/ in November 2018, project proponent needs to undertake inspection, testing, and calibration of the Metering System upon installation under observation of BPDB (Bangladesh Power Development Board). Thereafter, BPDB shall test the Metering System at intervals of not less than 180 (one hundred eighty) Days.

Personnel training:

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan. The CTI validation & verification team confirmed the above by reviewing the monitoring manual set by PP/23/ and confirmed the implementation of the training through the training records.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION OPINION

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

- The projects description compliance with, the requirements of Article 12 of the Kyoto Protocol/06/, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide, v4.4 and standard version 4.5/09/.
- The project's baseline and additionality are assessed against "ACM0002 Grid-connected electricity generation from renewable sources ---Version 21.0. and Tool for the demonstration and assessment of additionality – Version 07.0.0./09/.

•The project's monitoring plan is assessed against "ACM0002 Grid-connected electricity generation from renewable sources --Version 21.0. /09/

•A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

The project is expected to generate 649,135 tCO₂e during the length of entire crediting period (29-December-2021 to 28-December-2028).

Validated GHG emission reductions and removals in the above period:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1 (29-December-2021 to 31-December-2021) (3 days)	784
Year 1 (01-January-2022 to 31-December-2022)	95,449
Year 2 (01-January-2023 to 31-December-2023)	93,551
Year 3 (01-January-2024 to 31-December-2024)	93,037
Year 4 (01-January-2025 to 31-December-2025)	92,525
Year 5 (01-January-2026 to 31-December-2026)	92,016
Year 6 (01-January-2027 to 31-December-2027)	91,510
Year 7 (01-January-2028 to 28-December-2028) (362 days)	90,263
Total estimated ERs	649,135
Total number of crediting years	07
Average annual ERs	92,734

APPENDIX I: DOCUMENTS REVIEW UNDER VALIDATION

No.	Author	Title	References to the document	Provider
1.	Bangladesh Power Development Board	Commissioning certificate	Issued Dated: 21-December-2021	Project participant
2.	Applus	Contract of the PP with the DOE	Contract of the PP with the DOE	Project participant
3.	NA	Technical specifications	Technical specifications of all equipment from Manufacturers brochure.	Project participant
4.	PP	VCS PD Version 01 VCS PD version 02 VCS PD version 03 VCS PD version 04	11-May-2023 08-December-2023 25-December-2023 18-June-2024 (updated for VERRA PRR round 1)	Project participant
5.	PP	Emission Calculation Sheet- version 1.0- Estimated ER Sheet Final ER sheet 2.0 IRR spreadsheet, V1.0 IRR spreadsheet, V2.0 BM(Build Margin)OM	11-May-2022 08-December-2023 11-May-2022 08-December-2023 -	Project participant

		(Operating Margin)_ calculation Sheet Measured Efficiency(Confidential)	-	
6.	PP	- CDM validation and verification standard for project activities, Version 03.0 - Kyoto Protocol (1997)	https://cdm.unfccc.int/Reference/Standards/index.html https://unfccc.int/process-and-meetings/the-kyoto-protocol/what-is-the-kyoto-protocol/kyoto-protocol-targets-for-the-first-commitment-period	Project participant
7.	NA	Single Line Diagram	Plant Single Line Diagram	Project participant
8.	UNFCCC VERRA	Tools/ guidelines used in the project activity	UNFCCC CDM web site - UNFCCC Methodology: ACM0002 Grid- connected electricity generation from renewable sources —Version 21.0. (https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL010U2RYC6N3HASIXV7K84QBG9) - Tool 01 - Tool for the demonstration and assessment of additionality – Version 07.0.0 - Tool 07 - Tool to calculate the emission factor for an electricity system -Version 07.0 - Tool 24 – Common practice - Version 03.1 - Glossary of CDM terms version 11.0 - VCS standard V4.5 - VCS Program Guide V4.4	

09.	PP	• PPA • EPC Agreement • Module supply agreement	Power purchase agreement signed between Energon Renewables (BD) Ltd. and Bangladesh Power Development Board (BPDB) dated on 25-February -2019 EPC agreement signed between Energon Renewables (BD) Ltd. and ABB PTE LTD. dated 13-July-2020 Between PP and LONGi Solar Technology Co. Ltd dated 10/05/2021	PP

10.	PP	- Feasibility /DPR report - ERBL Information Memorandum - PV syst Report - Declaration of Annual O_M Cost - Loan sanction letter	Prepared by third Sgurr energy India Pvt. Ltd. Dated Internal document prepared by ERBL in 2020 Third party report prepared by Sgurr Energy Pvt. Ltd. ERBL dated 21-November-2023 Approved by Infrastructure Development Company Limited dated 8-February-2022	PP
11.	PP	LSC Document	Local stakeholders' consultation supporting: Notices, MoM's & photographs Feedback register etc.	Project participant
13.	PP	Declaration letter	For no double counting and part into any emission trading program and not received credits under another registry	Project participant
14.	PP	Local Stakeholder consultation	Snapshots and supportive documents, Feedback Forms, Meeting announcement documents etc. ESIA consultant Report	Project participant
15.	PP	Module supply agreement Agreement	Ref: ERBL_MSA_2021430 Dated: 10-May-2021	Project participant

16.	PP	BPDB Sources/Regulatory links	• notification(http://www.sreda.gov.bd/d3pbs_uploads/files/policy_1_rep_english.pdf) • http://bdlaws.minlaw.gov.bd/act-details-1228.html • http://www.sreda.gov.bd/d3pbs_uploads/files/policy_1_rep_english.pdf • https://www.bpdb.gov.bd/bpdb_new/d3pbs_uploads/files/11%20March%2019/Electricity%20Act%20(Gazette).pdf • https://bpdb.portal.gov.bd/sites/default/files/files/bpdb.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-11-29-05-22-0dea17e09d8a84e72a63312df6b5bdc6.pdf • https://www.bpdb.gov.bd/bpdb_new/d3pbs_uploads/files/11%20March%2019/Electricity%20Act%20(Gazette).pdf	Project participant
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APPENDIX II: CLARIFICATION REQUESTS, CORRECTIVE ACTION

CL ID	01	Section no.	3.1	Date: 24-November-2023
Description of CL				
PP requested to submit the following documents to the assessment team. 1. Technical specification of solar modules 2. Project ownership document 3. Operation & Maintenance certificate and cost. 4. Plant Layout of project activity				
Project participant response				Date : 08/12/2023
1. Technical specification of solar module enclosed. 2. Commissioning certificate along with PPA enclosed. 3. Operation and Maintenance has been done by the project owner. Plant declaration against O&M has been enclosed. 4. Single line diagram enclosed.				
Documentation provided by Project participant				
Technical Specification documents Commissioning certificate PPA O&M declaration letter Single line diagram				
VVB assessment				Date: 12-December-2023
PP has submitted the following documents to VVB team. 1. Technical brochure for the solar module o make Longi and technical details in revised PD are consistent with the document. 2. PP has submitted the Power purchase agreement and commissioning certificate of project activity from which it has been that the legal owner of the project activity is Energon Renewables (BD) Ltd. 3. Operation and maintenance done by project proponent and submitted declaration along with annual cost has been submitted by PP. 4. Single Line diagram has been provided by the PP same has been verified by VVB found inline with project activity. All above document as per requirement submitted by PP and found consistent and correct.				

Thus CL 01 is closed.

CL ID	02	Section no.	3.1	Date: 24-November-2023
Description of CL				
Information provided by PP during onsite visit, it has been observed that no. of modules in project activity differ from no. of modules provided in VCS PD. Kindly clarify.				
Project participant response				Date: 08-December-2023
The project activity involves installation of 248,136 solar PV module of different different module capacity. Typographical error in the PD has been updated suitably.				
Documentation provided by Project participant				
VVB assessment				Date: 12-December-2023
PP has now revised the number of the Solar module in VCS PD and VVB team cross verified with the Information memorandum document submitted by PP also confirmed during onsite visit and now found correct. Thus CL 02 is closed.				

CL ID	03	Section no.		Date: 24-November-2023
Description of CL				
PP requested to provide clarification on meter testing/calibration frequency as; during onsite visit interviews the frequency of the testing of the billing meters confirmed by PP as yearly by Bangladesh Power Development Board (BPDP). Further, BPDP conducts only testing of the both billing meters (main & check) and recommend for calibration if error found beyond acceptable limits. However, VCS PD mention calibration frequency in 180 days. kindly clarify.				
Project participant response				Date: 08-December-2023
Both the tariff meters for this project activity are under control of BPDB and both the meters under gone testing annually by the authority. If the error percentage exceeds the permissible error limit, BPDB advise the PP to conduct calibration. VCS PD has been updated suitably.				
Documentation provided by Project participant				
-				
VVB assessment				Date: 12-December-2023
PP has updated the information related to the meter calibration frequency in revised VCS PD calibration frequency is annual/yearly done on the basis same by BPDB, further VVB team has been confirmed same onsite. Thus CL 03 is closed.				

CL ID	04	Section no.		Date : 24-November-2023
Description of CL				

VCS PD is not clear about apportioning procedure if any applied for the mis-match between dates of the JMRs of first & last months and start & end dates of this monitoring period including vintage period breakups. PP requested to clarify the same.	
Project participant response	Date: 08-December-2023
Emission reduction calculation would be calculated based on the JMR. However, PP used to monitor and logged the daily generation and daily net grid export for the project activity. In the event of emission reduction calculation for any particular date or period which is less than a full month, logbook data for net grid export would be considered.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-December-2023
PP has revised the VCS PD now information updated in 5.3 of the PD. Thus, emission reduction calculation for any particular date or period can be calculated from logbook data for net grid export. Thus, acceptable to VVB team CL 04 is closed.	

CL ID	05	Section no.		Date: 24-November-2023
Description of CL				
During GSC period following period below comment has been raised. I don't think the additionality is demonstrated, and the "Tool for the demonstration and assessment of additionality" is used correctly, as the authors identify a single comparable project by using overly narrow search parameters. PP requested to clarify the same.				
Project participant response				Date: 08-December-2023
As per the CDM tool for common practice analysis (Tool 24 v3.1), the following conditions should be considered while doing the common practice analysis, - Applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity – Here the capacity of proposed project activity is 100 MW. So applicable range for the analysis is 50 MW to 150 MW. - The projects are located in the applicable geographical area – For this assessment, the applicable geographical area is Bangladesh - The projects apply the same measure as the proposed project activity – only projects applying the same measures as proposed project activity are considered - The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity – Here the source/fuel is solar energy.				

- e) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant – Here projects generating electricity through use of solar energy is considered
- f) As per the methodological tool the projects that have started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, are to be selected for consideration. The start date mentioned in PD is 13/07/2020 (Date of EPC contract signing) that resembles to start date definition of CDM project activity which is “the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system), for the CDM project activity or CPA. Where a contract is signed for such expenditures. – The projects which started commercial operations before 13th July 2020 are considered as per CDM.
- g) Identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation – Projects registered and/or submitted for registration and/or undergoing validation under any GHG abatement mechanism are excluded.

Only one project was identified with capacity of 50 MW and that too is registered under VERRA. So, that too was excluded which resulted in N_{all} to be 0. Please refer to below reports from Bangladesh Power Development Board for list of installed projects in Bangladesh till 2022.

1. https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-55-f1971a327c2aebfd37f6f9a8e723d1fb.pdf

2. https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-11-29-05-22-0dea17e09d8a84e72a63312df6b5bdc6.pdf

Hence, it could be concluded that the Tool for demonstration and assessment of additionality which refers to the common practice analysis tool has been correctly used to assess the project activity

Documentation provided by project participant

PP has updated the information in revised VCS PD. VVB team has checked that Tool for the demonstration and assessment of additionality i.e. TOOL 24 incorporated and common practice analysis also provided in VCS PD, Same has been verified with Link in response given for all project registered in Bangladesh no project registered in range 50 to 100 as per tool 24 requirement before 2020 (which is start date as per CDM definition first purchase order placed). Thus, acceptable to VVB team CL 05 is closed.

Table 2. CARs from this Project Validation

CAR ID	01	Section no.	3.1	Date:	24-November-2023
Description of CAR					

1. PP requested to provide DC capacity of Project in section 1.11 of PD. 2. LOGO provided on title page of the VCS PD for document prepared entity is not correct 3. PP has not provided weblinks or supporting document for laws mentioned in section 1.14, further also elaborate the laws and regulation. 4. Geo-coordinate of project activity found inline with the actual site, however as per provide the KML file for project activity as VCS standard requirement. 5. During review of the VCS PD, ER sheet & IRR sheet various inconsistencies were observed and same are commented in respective file/sections. Corrective actions sought for the same. Corrective action sought.	
Project participant response	Date: 08-December-2023
1. DC capacity has been mentioned in the section 1.11 of the VCS PD. 2. Updated 3. Section 1.14 has been elaborated suitably. 4. KML enclosed. 5. Inconsistencies are being updated in relevant places.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-December-2023
1. PP has updated DC capacity of Project activity in section 1.11 of revised VCS PD. 2. LOGO is now corrected in revised in title page revised VCS PD. 3. All laws and regulation link now incorporated in section 1.14 of revised VC PD. 4. PP has now submitted the KML file to VVB team and same has checked location found correct. 5. PP has now updated the VCS PD, IRR sheet and ER sheet. Thus, acceptable to VVB team. CAR 01 is closed.	

CAR ID	02	Section no.	3.1	Date: 24-November-2023
Description of CAR				
1. PP requested to update the VCS PD version 01 as per latest available VCS Standard Version 4.5. 2. Language of VCS PD is not appropriate in section 1.1 as project is already implemented and contributing towards emission reduction. 3. PP requested to provide complete year of crediting period in section 1.9 of VCS PD. 4. Section 1.10 of VCS PD mentions large project however, it is found not inline with the requirement of para 3.10.1 of VCS Standard 4.5. 5. Section 1.15 VCS PD mentions project not registered under any other mechanism or seeking registration, however supporting/weblinks is missing to justify this statement. Corrective action sought.				
Project participant response				Date: 08-December-2023

1. Updated 2. Section updated 3. Complete crediting period mentioned in section 1.9 of the VCS PD. 4. Section updated suitably 5. Plant declaration enclosed.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-December-2023
1. PP has updated the VCS PD version 01 as per latest requirement VCS Standard Version 4.5. 2. Language is now revised in section 1.1 of revised VCS PD as project is already implemented and contributing towards emission reduction. 3. PP has updated complete year of crediting period in section 1.9 section of revised VCS PD. 4. PP has updated section 1.10 of VCS PD now it is inline with the requirement of para 3.10.1 of VCS standard 4.5 5. PP has provided the declaration letter dated project is not registered under any other mechanism or seeking registration. Thus, acceptable to VVB team. CAR 02 is closed.	

CAR ID	03	Section no.	3.2.2	Date: 24-November-2023
Description of CAR				
Section 2.2 of the VCS PD is not inline with the VCS MR template guideline version 4.2, further PP is requested to provide supporting documentation regarding local stakeholder involvement before the implementation of project activities, as mandated by the requirements of Paragraph 3.18.2 of VCS Standard Version 4.5. Kindly submit supporting and revised VCS PD.				
Project participant response				Date: 08-December-2023
Section 2.2 has been updated suitably. Relevant supporting documents enclosed.				
Documentation provided by Project participant				
-				
VVB assessment				Date: 12-December-2023
PP has revised section 2.2 of VCS PD as per template and further provided the all supporting related to the project activity for local stakeholder involvement before the implementation of project activities as per requirement of Paragraph 3.18.2 of VCS Standard Version 4.5. Thus CAR 03 is closed.				

CAR ID	04	Section no.	3.3.5	Date: 24-November-2023
Description of CAR				

1. In section 3.5 of VCS PD complete information for investment analysis accordance with tool 27 is missing. 2. Information related to decision making of the project activity in VCS PD and the supporting for same missing in document. 3. PP requested to submit the supporting document for below: - i) Actual PLF of Project activity. ii) Actual total Cost of Project activity (Including land cost). iii) Purchase orders and O&M cost expenses. iv) Invoice and JMR copies. Corrective action sought.	
Project participant response	Date: 08-December-2023
1. Section 3.5 has been updated suitably and investment analysis as per tool 27 has been incorporated. 2. Information provided. 3. (i) Annual average of 20.29% from December -22 to November-23. Supporting Generation details is attached. (ii) Purchase order enclosed. (iii) Purchase order and declaration against O&M expenses enclosed. (iv) Copy enclosed	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-December-2023
1. PP has now included the complete information for investment analysis accordance with tool 27 in VCS PD. Thus, acceptable to assessment team. 2. PP has now updated information related to decision making for project activity date of first purchase has been taken as investment date for the project activity and evidence also submitted by PP. 3. i) Actual sheet for the PLF achieved by project activity during previous year (2022-2023) has been submitted by PP. ii) PP has provided all purchase order. iii) PP has provided all purchase order and O&M declaration of the project activity. iv) JMR and Invoices sample copies have been submitted by PP. All above documents found consistent with project activity. Thus CAR 04 is closed.	

CAR ID	05	Section no.	3.3.5	Date: 24-November-2023
Description of CAR				
During desk review of IRR sheet following point observed by assessment team: - 1. In tab Benchmark, Source link of "Inflation forecast (Average consumer price) as per IMF for 5 years", 6.48% inflation is mentioned nowhere. 2. In tab P&L Statement, Land cost is treated as an expense. Land cost cannot be the part of revenue Expenses.				

3. In tab P&L Statement, Residual Value added in cash inflows in cell AB52 is Net Depreciable Value instead of Salvage value. (22.15 should be added instead of 77.06)
4. In tab Assumption, Term loan interest rate is "as on 02-June-2023" which is after the date of investment decision.
5. In tab P&L Statement, while calculating Tax Payable, Book depreciation is considered instead of depreciation as per income tax act. Also, if there is any difference in Book Depreciation and IT Depreciation, Tax shield should be considered.
6. Date on PVsyst report is 21-July-2021 which is after the date of Investment Decision. Hence PVsyst Report is not considered.
7. Annual generation mentioned in project document is 179,998 MWh on page 4. Annual Generation mentioned in IRR sheet is 1,75,648.78. There is a mismatch in both.
8. Detailed project report, PPA and various documents mentioned in source column were not provided, hence some assumptions could not be cross verified. 10. sources of financial parameters, Margin %, Land cost, salvage value is not mentioned in IRR sheet.

Corrective action sought.

Project participant response		Date: 08-December-2023
1. Source has been updated in the IRR sheet. Corresponding value is also updated. 2. P&L statement section updated 3. Cell AB32 in P&L statement tab has been updated 4. Term Loan interest rate has been considered as per the information memorandum. IRR sheet has been updated accordingly. 5. Section updated 6. Previous PV Stys report enclosed. 7. Section updated 8. Source of information provided in IRR sheet 9. Source provided		
Documentation provided by project participant		
-		
VVB assessment		Date: 12-December-2023
1. PP has updated the Inflation forecast (Average consumer price) as per IMF for 5 years) and also Link for same is provided. 2. P&L table is now updated in IRR sheet Land cost is now removed. 3. Salvage Value is now updated in revised IRR sheet in cell AB 52. thus, acceptable to assessment team. 4. Term loan is updated as per investment decision date. Thus, acceptable to VVB team. 5. Tax rate is now updated in sheet as per Book depreciation. 6. Correct PV syst report now submitted by PP. 7. PP has updated value in revised VCS PD for Annual generation now value is consistent with IRR sheet. 8. Source for all parameters are now provided in Assumption tab of IRR sheet and supporting also submitted for the same. Thus, acceptable to VVB team CAR 05 is closed.		

CAR ID	06	Section no.	3.3.6	Date: 24-November-2023
Description of CAR				

Estimated emission reduction sheet does not contain the reference link/ data source for the combined emission factor calculations. Further value for combined emission factor value is not inline with the VCS PD. Corrective action sought.	
Project participant response	Date: 08-December-2023
Updated grid emission factor calculation sheet enclosed.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-December-2023
PP has submitted updated the grid emission factor and provided link/data source for the calculation of combined emission factor is now provided in revised VCS PD further IRR sheet has been also submitted. Thus, CAR 06 is closed.	

Table 2. FARs from this Project validation

FAR ID	xx	Section no.	Date: DD/MM/YYYY
Description of FAR			
Project participant response			Date: DD/MM/YYYY
Documentation provided by project participant			
VVB assessment			Date: DD/MM/YYYY

APPENDIX III: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Validation Team Member: -

No	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interview(s)	Validation findings
1.	Lead Auditor/ Technical Expert	OR	Takarkhede	Dr. Atul	TQC- Outsourced entity	Yes	Yes	Yes	Yes

Technical Reviewer and Approver of the Validation Certification Report: -

No.	Role	Type of resource	Last name	First name	Affiliation (e.g., name of central or other office of DOE or outsourced entity)
1.	Technical reviewer (TR)	EI	Singh	Dr. N Premjit	Applus+ Certification
2.	Approver	IR	Calle de Miguel	Agustin	Applus+ Certification

Shorts CV of the Team:

- Dr. Atul Takarkhede** is Ph.D. (Environmental Sciences) from Institute of Science, RTM Nagpur University, Nagpur, and he has already published different technical papers related to environmental sciences. He counts with more than 15 years of experience in field of environmental Auditing, consulting and accreditation. He is an expert in ISO 9001-14001, CO2/GHG Reporting, Carbon Foot Print, Energy, Water and Waste Management reporting for organizations' environmental performance. His professional portfolio is mainly related with carrying out EIA, conducting QA/QC of EIA Reports; conducting environmental/water audits;

NABET requirements appliance, functional area expert in Water Pollution & Solid & Hazardous Waste management among others. Furthermore, he counts with solid experience on CDM-VCS-GS consultancy and auditing. Currently he is associated with True Quality Certifications Private Limited and empanelled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Dr. Atul Takarkhede is based in Nagpur, India.

- **Dr. N Premjit Singh** has a PhD in Mechanical Engineering (Thesis: Design and development of a square parabolic dish system with a concentrated photovoltaic (CPV) module for performance improvement) from the Indian Institute of Technology (IIT) Madras, Chennai, India, awarded in 2021. M.Tech in Energy Technology, Tezpur University, Napaam, India (2007), and B.Tech in Mechanical Engineering (2005), NERIST, Nirjuli, India. He has extensive experience of about 7 years with DOEs, including UNFCCC CDM and other carbon related schemes (e.g., VCS, GS, GCC), and 5 years + in research projects, renewable energy, and energy audits. In Applus+ since March 2023, he has been the Product Assurance Manager for CDM/VCS/GS4GG/GCC Department to ensure the quality of the performance of different assessments. Coordinate the global team for technical reviews, and identify the training needs for the auditors and technical reviewers to improve the quality of reports. Holds experience as a Lead Auditor, Validator and Verifier for GHG mitigation projects and programmes of activities in Sectoral Scope 1.2 (Renewables), and is qualified as per Applus+ procedures. Dr. N Premjit Singh is based in Gurugram, India. Dr. N Premjit Singh participates as part of Technical Reviewer team.

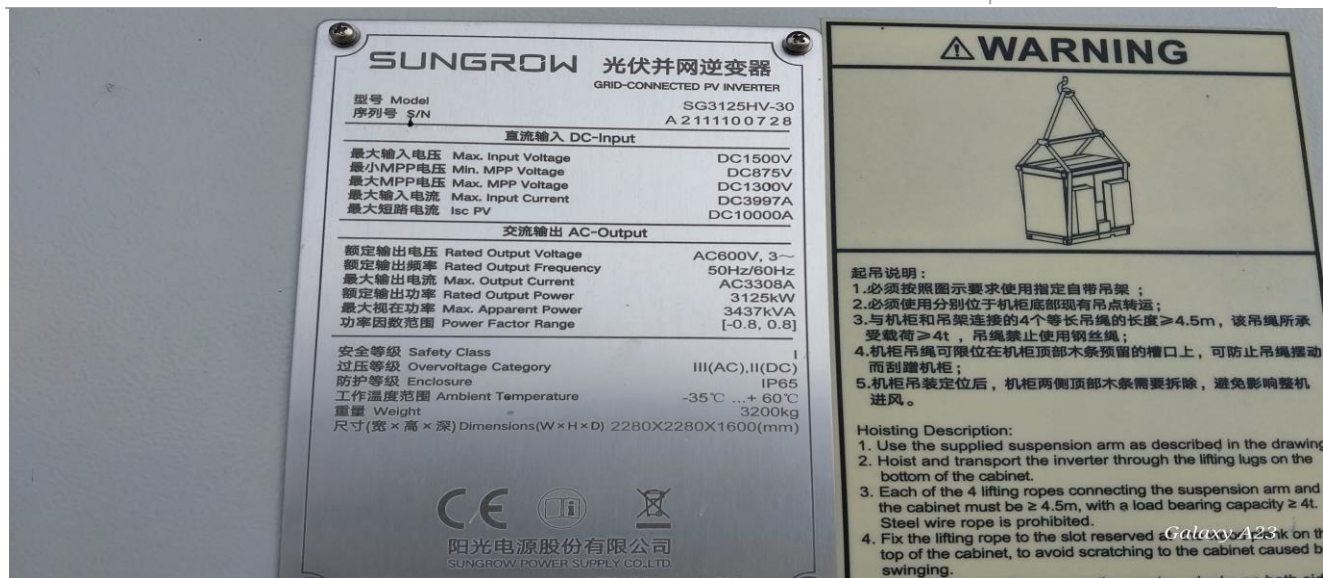
APPENDIX IV: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CEA	Central Electricity Authority
CL	Clarification request
CERC	Central Electricity Regularity Commission
CMS	Central Monitoring system
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DOE	Designated Operational Entity
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
OM	Operating Margin
GWP	Global Warming potential
JMR	Joint Metering reading
VVB	Validation/Verification Body

APPENDIX V: ONSITE EQUIPMENTS AND NAMEPLATE PHOTOGRAPHS



I) ENERGY METERS AT THE SITE



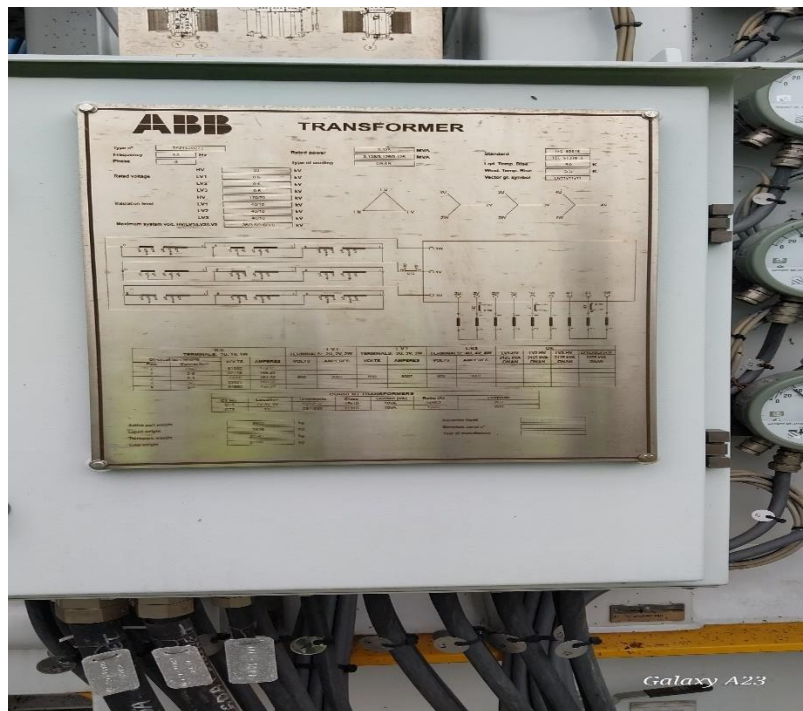
II) INVERTOR SPECIFICATION



III) POWER TRANSFORMERS



IV) MODULE SOLAR PLANT



V) TRANSFORMER



VI) SOLAR PLANT